AGILE SPRINT PLANNING &
USER STORY WORKSHOP

Intelligent Water Distribution Monitoring and Leakage Detection Platform

CO1 — AT-2: Software Engineering
Comprehensive Sprint Planning Document

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1. Introduction & Problem Statement

Municipal water utilities lose a significant share of treated water — commonly 20% to 40% globally — before it ever reaches a paying customer. This “Non-Revenue Water” (NRW) results primarily from undetected pipe leaks, illegal connections, and metering inaccuracies. Traditional leak detection relies on manual inspection, customer complaints, or after-the-fact billing anomalies, all of which are slow, reactive, and expensive.

The Intelligent Water Distribution Monitoring and Leakage Detection Platform is a software system that combines IoT sensor networks (flow meters, pressure transducers, and acoustic leak sensors), a geospatial digital twin of the pipeline network, and machine-learning analytics to detect leaks and anomalies in near real time. The platform notifies operations staff and field technicians automatically, shortens response time from days to minutes, and gives utility managers data-driven tools to plan preventive maintenance and reduce water loss.

This document applies Agile software engineering practices — vision framing, stakeholder analysis, epics and user stories, acceptance criteria, backlog prioritization, sprint planning, story-point estimation, and sprint-goal definition — to translate this problem statement into an actionable, developer-ready plan for the first development sprint and the sprints that follow.

2. Project Vision and Stakeholders

2.1 Project Vision

> ***To build a scalable, real-time platform that reduces Non-Revenue Water by at least 25% within the first year of deployment, by detecting pipeline leaks within minutes of onset, visualizing the distribution network spatially, and empowering operators, field crews, and consumers with timely, actionable information — ultimately conserving water resources and improving service reliability for the community.***

2.2 Objectives

- Ingest and visualize live flow, pressure, and acoustic sensor data across the distribution network.
- Automatically detect leaks, bursts, and abnormal consumption patterns using rule-based and ML-based analytics.
- Reduce mean-time-to-detect (MTTD) leaks from days to under 15 minutes, and mean-time-to-repair (MTTR) by at least 30%.
- Provide a GIS-based digital twin of pipes, valves, and reservoirs for spatial decision-making.
- Enable field technicians to receive, navigate to, and close work orders from a mobile application.
- Give consumers visibility into their consumption and proactive outage notifications.
- Support regulatory compliance reporting on water loss and service quality.

2.3 Key Stakeholders

Stakeholder	Role / Interest
Municipal Water Utility (Client/PO org.)	Owns the platform; defines strategic NRW-reduction goals and funds the project.
Water Utility Operations Manager	Oversees daily operations; primary decision-maker on alert thresholds and SLAs.
Field Maintenance Technicians	Respond to leak alerts, perform repairs, update work-order status via mobile app.
SCADA / IoT Sensor Engineers	Install, calibrate, and maintain flow meters, pressure and acoustic sensors.
Network Operations Center (NOC) Staff	Monitor real-time dashboards, triage alerts, escalate critical incidents.
GIS / Asset Management Team	Maintains the digital twin of the pipeline network, valves, and reservoirs.
Data Analytics / ML Engineers	Build and maintain leak-prediction and consumption-forecasting models.
Municipal Government / Regulatory Body	Requires compliance reporting on water loss, quality, and conservation targets.
End Consumers (Residential/Commercial)	Benefit from fewer outages and receive usage insights via portal.
Billing & Customer Service Dept.	Uses consumption data for accurate billing and dispute resolution.
IT/Cybersecurity Team	Ensures data security, uptime, and compliance with critical-infrastructure standards.
Scrum Master / Development Team	Facilitates Agile ceremonies; cross-functional team builds the platform.

3. Epics

The product backlog is organized into seven epics, each representing a large body of related functionality delivered incrementally across multiple sprints.

ID	Epic Name	Description
EPIC-1	Real-Time Sensor Data Acquisition & Monitoring	Ingest, process, and visualize live flow, pressure, and quality data from IoT sensors.
EPIC-2	Leak & Anomaly Detection Engine	Detect leaks, bursts, and abnormal consumption using rule-based and ML models; raise alerts.
EPIC-3	GIS Network Visualization & Asset Mgmt.	Maintain a geospatial digital twin of pipelines, valves, reservoirs, and sensor locations.
EPIC-4	Predictive Maintenance & Analytics	Forecasting, historical trend analysis, and maintenance scheduling to reduce NRW.
EPIC-5	Field Technician Mobile Application	Receive work orders, navigate to leak sites, and log repair outcomes from the field.
EPIC-6	Consumer Engagement Portal	View consumption, receive outage notifications, and report issues.
EPIC-7	System Admin, Security & Reporting	Manage users/roles, secure access, and generate regulatory/compliance reports.

4. User Stories & Acceptance Criteria

Each story follows the standard Agile format (“As a <user>, I want <feature>, so that <benefit>”), with MoSCoW priority, business value, story points, and target sprint. Acceptance criteria are summarized in Given–When–Then form.

EPIC-1 Real-Time Sensor Data Acquisition & Monitoring

US-01 *As a NOC operator, I want to see live flow and pressure readings on a dashboard, so that I can monitor network health in real time.*

Priority: Must Value: 5/5 Points: 8 Sprint: 1

Acceptance: Given a sensor is transmitting, when a new reading arrives, then the gauge updates within 5s; a sensor silent for 2+ minutes is marked ‘offline’ and logged.

US-02 *As a SCADA engineer, I want to register and configure new IoT sensors through an admin interface, so that new deployment points are onboarded without code changes.*

Priority: Must Value: 4/5 Points: 5 Sprint: 1

Acceptance: Given a SCADA engineer submits a sensor's ID, type, and GPS coordinates, then it appears in the device registry and begins accepting telemetry.

US-03 *As a data engineer, I want raw sensor telemetry buffered and stored in a time-series database, so that historical analysis and audits are possible.*

Priority: Must Value: 4/5 Points: 5 Sprint: 2

Acceptance: Given telemetry received via MQTT, then it is persisted with device ID and timestamp within 2 seconds.

EPIC-2 Leak & Anomaly Detection Engine

US-04 *As an operations manager, I want the system to automatically flag a suspected leak when flow exceeds the expected consumption baseline, so that response time is minimised.*

Priority: Must Value: 5/5 Points: 13 Sprint: 1

Acceptance: Given an established night-flow baseline, when flow exceeds threshold for 15 continuous minutes, then a ‘High’ severity alert is raised and escalates via SMS/email if unacknowledged within 10 minutes.

US-05 *As a data analyst, I want a machine-learning model that classifies acoustic sensor signatures, so that leaks are distinguished from normal operational noise with high confidence.*

Priority: Should Value: 4/5 Points: 13 Sprint: 3

Acceptance: Given streamed frequency data, when the model scores a segment, then a leak probability above 85% creates a candidate leak event for review.

US-06 *As a NOC operator, I want to acknowledge, prioritize, and close alerts from a unified queue, so that duplicate work orders are avoided.*

Priority: Must Value: 4/5 Points: 5 Sprint: 2

Acceptance: Given overlapping alerts, when one is acknowledged, then related alerts group under a single incident and the queue updates in real time.

EPIC-3 GIS Network Visualization & Asset Management

US-07 *As a GIS analyst, I want to view the pipeline network, valves, and sensor locations on an interactive map, so that I can spatially correlate alerts with infrastructure.*

Priority: Must Value: 5/5 Points: 8 Sprint: 1

Acceptance: Given the asset DB contains pipe/valve geometries, then assets render as layered map objects, zoomable/pannable and filterable by type.

US-08 *As a GIS analyst, I want a leak alert to be automatically plotted on the map at its estimated location, so that field crews can be dispatched precisely.*

Priority: Must Value: 5/5 Points: 8 Sprint: 2

Acceptance: Given an alert with sensor GPS references, when triangulated between the two nearest sensors, then a severity-coded marker appears within 30 seconds.

EPIC-4 Predictive Maintenance & Analytics Dashboard

US-09 *As an operations manager, I want a dashboard showing Non-Revenue Water (NRW) trends over time, so that I can measure the impact of leak-reduction efforts.*

Priority: Should Value: 4/5 Points: 8 Sprint: 3

Acceptance: Given 30+ days of data, then the dashboard shows daily/weekly/monthly NRW % with trend line and CSV export.

US-10 *As a maintenance planner, I want the system to recommend a pipe-replacement priority list based on age, material, and burst history, so that capital repair budgets are optimally allocated.*

Priority: Could Value: 3/5 Points: 13 Sprint: 4

Acceptance: Given asset age/material/burst records, when the weekly scoring job runs, then the top 20 highest-risk segments are ranked with a risk score.

EPIC-5 Field Technician Mobile Application

US-11 *As a field technician, I want to receive a push notification with leak location and severity, so that I can respond immediately without waiting at a desk.*

Priority: Must Value: 5/5 Points: 5 Sprint: 2

Acceptance: Given an alert is assigned, then the technician's app receives a push notification with map coordinates and severity within 10 seconds.

US-12 *As a field technician, I want turn-by-turn navigation to the leak site from within the app, so that I do not need a separate mapping tool.*

Priority: Should   Value: 3/5   Points: 3   Sprint: 3

Acceptance: Given a work order has GPS coordinates, when the technician taps ‘Navigate’, then the device's native navigation view opens with the destination pre-filled.

US-13 *As a field technician, I want to log repair details, photos, and parts used after completing a job, so that maintenance history is accurately recorded.*

Priority: Must   Value: 4/5   Points: 5   Sprint: 2

Acceptance: Given a completed repair, when the closure form with notes and a photo is submitted, then status changes to ‘Resolved’ and syncs centrally.

EPIC-6 Consumer Engagement Portal

US-14 *As a residential consumer, I want to view my daily water consumption on a web/mobile portal, so that I can monitor and reduce my usage.*

Priority: Should   Value: 3/5   Points: 5   Sprint: 3

Acceptance: Given a linked smart meter, then the portal shows a 30-day daily consumption chart with locality comparison.

US-15 *As a consumer, I want to receive an outage or maintenance notification for my area, so that I can plan around service interruptions.*

Priority: Could   Value: 3/5   Points: 5   Sprint: 4

Acceptance: Given a planned maintenance or active leak affecting a zone, then registered consumers receive an SMS/app notice with estimated restoration time.

EPIC-7 System Administration, Security & Reporting

US-16 *As a system administrator, I want to create role-based accounts (Admin, Operator, Technician, Analyst, Consumer), so that access to sensitive functions is properly restricted.*

Priority: Must   Value: 4/5   Points: 5   Sprint: 1

Acceptance: Given a role is assigned at creation, when the user logs in, then the UI exposes only permitted modules and actions.

US-17 *As a compliance officer, I want to generate a monthly regulatory report on water loss and quality incidents, so that the utility meets government reporting obligations.*

Priority: Should   Value: 3/5   Points: 8   Sprint: 4

Acceptance: Given a calendar month has closed, then a PDF summarizing NRW %, incident count, and resolution times is generated within 60 seconds.

5. Product Backlog Prioritization

The backlog is prioritized using MoSCoW classification and business-value scoring (1–5), informed by mission impact, technical dependencies (ingestion → detection → GIS overlay), and stakeholder urgency. Foundational 'Must' stories that unblock later work are scheduled first even where direct value is moderate.

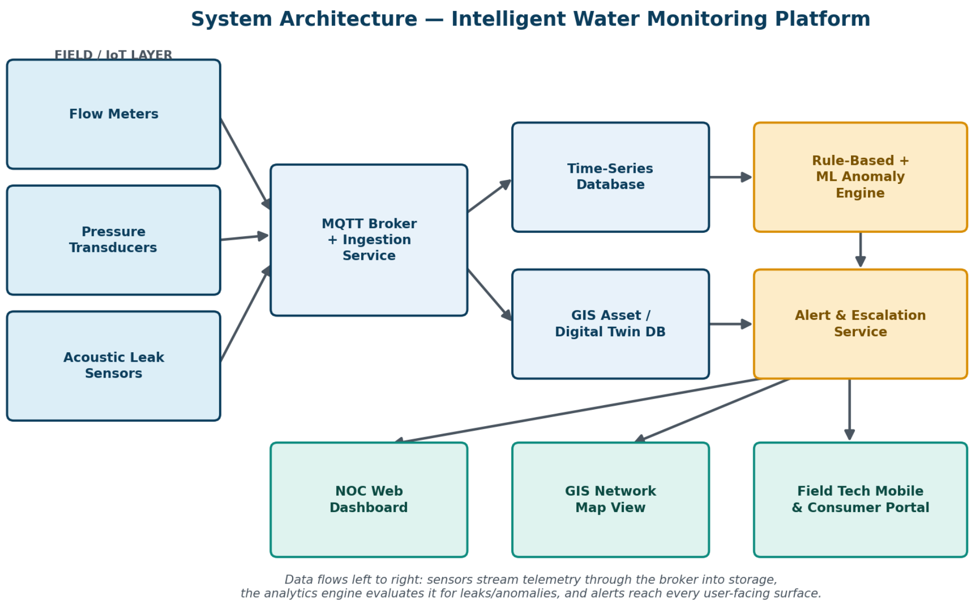

Figure 1 — Platform system architecture: sensor layer through ingestion, storage, analytics, and user-facing surfaces.

ID	Story (summary)	MoSCoW	Value	Pts	Sprint
US-01	Live flow/pressure dashboard	Must	5	8	Sprint 1
US-04	Auto leak-flag on baseline breach	Must	5	13	Sprint 1
US-07	Interactive GIS network map	Must	5	8	Sprint 1
US-08	Auto-plot leak alert on map	Must	5	8	Sprint 2
US-11	Push notification to technician	Must	5	5	Sprint 2
US-02	Sensor onboarding admin UI	Must	4	5	Sprint 1
US-03	Time-series telemetry storage	Must	4	5	Sprint 2

ID	Story (summary)	MoSCoW	Value	Pts	Sprint
US-06	Unified alert acknowledgement queue	Must	4	5	Sprint 2
US-13	Repair logging with photos	Must	4	5	Sprint 2
US-16	Role-based account creation	Must	4	5	Sprint 1
US-05	Acoustic ML leak classifier	Should	4	13	Sprint 3
US-09	NRW trend dashboard	Should	4	8	Sprint 3
US-12	In-app turn-by-turn navigation	Should	3	3	Sprint 3
US-14	Consumer consumption portal	Should	3	5	Sprint 3
US-17	Monthly regulatory report	Should	3	8	Sprint 4
US-10	Pipe-replacement priority model	Could	3	13	Sprint 4
US-15	Consumer outage notification	Could	3	5	Sprint 4

Dependency note: US-04 depends on US-01's telemetry baseline; US-08 depends on US-04 and US-07; US-11 depends on US-04. These dependencies drive the sprint sequencing above.

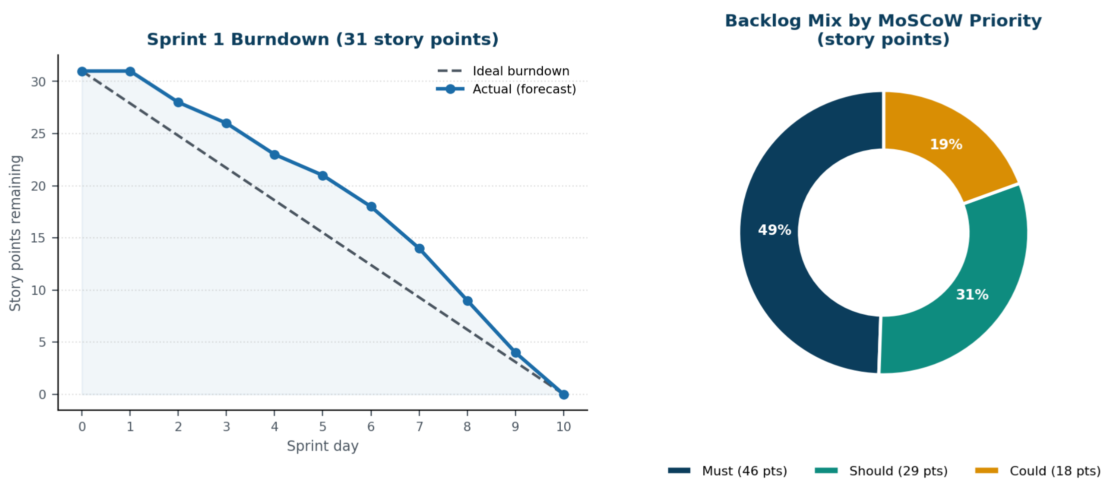

Figure 2 — Sprint 1 burndown forecast and backlog point-mix by MoSCoW priority.

6. Sprint Backlog — Sprint 1

Sprint 1 is a two-week sprint establishing the platform's foundation: live telemetry ingestion, the core leak-alerting rule engine, the base GIS map, sensor onboarding, and role-based access control.

6.1 Selected Stories

ID	Story	Points
US-01	Live flow/pressure dashboard for NOC operators	8
US-02	Sensor registration & onboarding admin UI	5
US-04	Automatic leak flag on baseline breach + escalation	13
US-07	Interactive GIS map of pipeline network	8
US-16	Role-based account creation (RBAC)	5

6.2 Task Breakdown (representative)

Story	Task	Owner	Hrs
US-01	Configure MQTT broker & build gauge widgets	Backend/Frontend Dev	16
US-02	Device registry schema + onboarding form + validation	Backend/Frontend/QA	12
US-04	Night-flow baseline job + rule engine + SMS/email escalation	Data/Backend Dev	24
US-04	Automated tests for alert-escalation timing	QA Engineer	5
US-07	GIS map component, asset geometries, type filtering	Frontend/Backend Dev	18
US-16	RBAC schema, admin screen, route/API guards	Backend/Frontend Dev	16

Total Sprint 1 capacity: a 5-person cross-functional team (2 backend, 1 frontend, 1 data engineer, 1 QA) at 6 productive hrs/day over a 10-day sprint ≈ 300 hours. Planned effort above totals ~95 hours, leaving buffer for review, ceremonies, and unplanned work — consistent with sustainable-pace Agile practice.

7. Story Point Estimation

Story points were assigned using Planning Poker with the modified Fibonacci sequence (1, 2, 3, 5, 8, 13, 21), calibrated against US-16 (RBAC) as a 5-point baseline. 1–2: trivial, no unknowns. 3: small, self-contained. 5: moderate, 2–3 components. 8: complex, cross-cutting/real-time. 13: high complexity/uncertainty (new algorithms, ML). 21: epic-sized — must be decomposed.

7.1 Estimation Rationale (Selected Examples)

- US-04 (leak alert engine) — 13 pts: statistical baseline model, rules engine, multi-channel escalation, timing tests across ingestion, backend, and notification services.
- US-01 (live dashboard) — 8 pts: real-time pipeline (MQTT → backend → WebSocket → UI) plus offline-sensor detection; moderate integration risk.
- US-02 (sensor onboarding) — 5 pts: standard CRUD with validation; low uncertainty once schema is defined.
- US-05 (acoustic ML classifier) — 13 pts: model training data, audio feature extraction, confidence scoring; high uncertainty until dataset is validated.

Team velocity from prior comparable sprints (or a conservative first-sprint forecast) is assumed at 30–35 points/two-week sprint; Sprint 1 commits to 31 points, within this range.

8. Sprint Goal and Expected Deliverables

8.1 Sprint Goal

> ***"Stand up the platform's real-time monitoring foundation — live sensor telemetry, automated leak alerting, a base GIS network map, and secure role-based access — so that a NOC operator can detect and view a simulated leak on the map within minutes of it occurring."***

8.2 / 8.3 Deliverables & Definition of Done

- Working telemetry pipeline with live dashboard and offline-sensor detection.
- Sensor/device registry with onboarding UI usable without developer involvement.
- Leak-alert rule engine flagging threshold breaches and escalating unacknowledged alerts.
- Interactive GIS base map with pipelines, valves, sensors, and type-based filtering.
- RBAC with distinct Admin, Operator, Technician, Analyst, and Consumer permissions.
- DoD: code merged and peer-reviewed, CI passing, acceptance criteria verified in staging, no critical/high defects, docs updated, story demoed and accepted at Sprint Review.

8.4 Looking Ahead

Sprint 2 persists historical telemetry, groups alerts into incidents, plots leaks geospatially, and delivers the first field-technician workflow. Sprint 3 introduces the acoustic ML classifier, in-app navigation, and the consumer portal. Sprint 4 rounds out predictive maintenance, consumer outage notifications, and compliance reporting — completing the MoSCoW Should/Could backlog from Section 5.